Seasonal and Weather-Based HSL City Bike Allocation

Comprehensive Empirical Findings, Leakage-Free Predictive Benchmarking & Rebalancing Triage

Course Project: Canvas Workshop 3 (Machine Learning in Operations) | Target: City-Bike Operators & HSL Transport Planners
Authors: Chenyu Wu, Jizhou Wang, Jussi Honkio | System Scope: Helsinki & Espoo Municipalities (457 Stations)
Temporal Coverage: 2 Full Cycling Seasons (2024 & 2025; April 1 ? October 31 each; 14 Operating Months Total; 4,951,245 Valid Trips)
Artifacts: Production Data Pipeline, Leakage-Free Models, GIS Triage Gallery & Interactive GitHub Pages Planning Platform

1. EXECUTIVE SUMMARY & CANVAS PROBLEM DEFINITION | Operational Fleet Management Context

Efficient urban bike-sharing in northern climates is governed by seasonal swings, subarctic weather transitions, and diurnal commuter tides. Grounded in the HSL City Bike Project Canvas, this study evaluates two full cycling seasons (2024?2025; 14 operating months) across Helsinki and Espoo to resolve fundamental fleet allocation challenges: where should bikes be concentrated, how reliably can station demand be forecast without data leakage, and which stations suffer from chronic stockouts or dock congestion.

Core Empirical Conclusions: (1) Out of 5.10M raw trips, 4,951,245 clean journeys (97.04% retention) were synthesized with official ArcGIS station coordinates and FMI ground-truth weather. (2) Daily ridership exhibits a sharp unimodal summer envelope and strong temperature correlation (r = +0.74), but temperature co-moves with daylight and holiday schedules, establishing descriptive rather than causal links. (3) Weekday demand features an evening rush surge (+70% to +105% over morning) and commuter tidal reversals between residential and business hubs. (4) An out-of-time predictive benchmark proves that a historical same-month baseline achieves lowest volume error (17.58% WAPE), while causal Ridge regression excels at early-season deficit rank ordering (up to 80% Top-20 match). (5) A zero-dependency GitHub Pages planning platform is deployed.

2. CORE RESEARCH QUESTIONS | Investigative Framework

Q1. Seasonal & Climatological Trajectories: How does station-level and municipal demand evolve across the 7-month operating window (April?October), and how strongly does daily trip volume co-move with ground-truth temperature and precipitation observations?

Q2. Intraday & Weekday/Weekend Regimes: How does demand diverge between workdays and weekends? To what extent do morning (07:00?09:00) and evening (16:00?18:00) peak rush hours exhibit directional volume asymmetries that monthly aggregates obscure?

Q3. Out-of-Time Demand Predictability: Can a historical same-month baseline (2024 -> 2025) reliably forecast departures and returns? How does it compare against causal, leakage-free Ridge regression and decision trees when strictly restricted to preceding-month lags?

Q4. Rebalancing Triage & Network Bottlenecks: Which stations exhibit persistent, severe directional flow imbalances (net_flow = returns - departures), and how should transport planners prioritize stations facing chronic bicycle depletion (sinks) versus dock saturation (sources)?

3. DATA INGESTION, HYGIENE FILTERS & UNMAPPED DOCKS | Data Governance & Traceability

HSL Origin-Destination (OD) Journey Records:
? Raw Ingested Trips: 5,102,233 records across 14 monthly CSVs from official HSL Open Data (2,585,668 in 2024; 2,516,565 in 2025).
? Valid Clean Records Retained: 4,951,245 trips (2,507,175 in 2024, 96.96% retention; 2,444,070 in 2025, 97.12% retention; Overall: 97.04%).

Strict 4-Part Data Hygiene Filtering Rules:
1. Trip Duration: 10s <= t <= 86,400s (24h). Discards accidental instant checkouts and unreturned/lost bicycles.
2. Trip Distance: Covered distance >= 10m. Eliminates stationary dock sensor testing and immediate re-docking.
3. Speed Boundary: Mean velocity <= 10.0 m/s (~36 km/h). Filters motorized transit boardings and GPS teleportation glitches.
4. Chronological Integrity: Return timestamp >= Departure timestamp.
? Midnight Crossing Attribution: 21,532 trips (0.86%) in 2024 and 19,957 trips (0.82%) in 2025 crossed midnight; departures are attributed to day D1 at origin, returns to D2 at destination.

Station GIS Metadata & Coordinate Alignment:
? Ingested from official HSL ArcGIS REST feature service covering 457 active stations with English/Finnish/Swedish names, capacity, and WGS84 coordinates.
? 454 stations have verified geographic coordinates and are mapped.

Audited Unmapped Stations (Zero Silent Imputation):
? 4 stations in 2024 (IDs 710, 768, 997, 999; 26 rows) and 7 in 2025 (IDs 626, 710, 716, 754, 768, 997, 999; 45 rows) represent maintenance workshops or pop-up test racks absent from the ArcGIS layer.
? Preserved as NaN for coordinates/municipality to avoid synthetic GIS distortion. Retained in system trip totals (23,981 departures in 2025).

FMI Retrospective Ground-Truth Weather Observations:
? Downloaded via FMI WFS API for Helsinki Kaisaniemi (FMISID 100971) and Espoo Tapiola (FMISID 874863). 428 calendar days per station across both seasons; 100% complete across mean, min, max temperature, precipitation, and snow depth. FMI '-1.0' code converted to 0.0 mm.

Operating-Day Calendar Normalization Standard: Each 214-day cycling season consists of exactly 154 weekdays and 60 weekend days. To eliminate calendar-composition bias (e.g., months with 5 weekends vs. 4), all intraday and peak-hour metrics throughout this report and interactive tools are reported as Operating-Day Normalized Daily Averages (trips / operating days), never as raw sum comparisons.

4. SEASONAL RIDERSHIP TRAJECTORY & MUNICIPAL DIVERGENCE | Macro Demand Patterns

TOTAL VALID RIDERSHIP	HELSINKI METROPOLITAN SHARE	ESPOO SUBURBAN SHARE
4,951,245 Trips	**88.8% Regional Share**	**11.2% Regional Share**
2024: 2.51M | 2025: 2.44M (-2.5% YoY)	2.20M (2024) | 2.17M (2025) Trips	294k (2024) | 260k (2025) Trips

? Unimodal Seasonal Envelope: Demand traces a pronounced summer unimodal bell curve governed by high-latitude Nordic daylight and climate. System ridership peaks in July and August (450,000?492,000 trips/month) and contracts steeply during the spring and autumn shoulder months (April: 162,000?243,000 trips; October: 198,000?207,000 trips).
? Municipal Structural Disparity: Helsinki commands nearly 90% of regional activity, supported by dense inner-city cycling grids, intermodal train hubs (Central Station, Pasila), and high docking density. Espoo commands 11.2%, anchored by West Metro feeder hubs (Matinkylä, Niittykumpu) and the Aalto University Otaniemi campus. Espoo trips exhibit higher leisure elasticity and lower absolute shoulder resilience.

5. FMI WEATHER OBSERVATIONS & CONFOUNDER DISCLAIMERS | Retrospective Climatological Correlations

Empirical Ground-Truth Correlation Analysis (214 daily observations in 2025 across Helsinki & Espoo):
? Temperature Sensitivity: Daily trips exhibit strong positive linear correlation with mean daily temperature (r = +0.737 in Helsinki, r = +0.665 in Espoo) and maximum daily temperature (r = +0.788 in Helsinki, r = +0.751 in Espoo). Ridership rises continuously from 5°C to 25°C.
? Precipitation Impact: Daily volume displays a moderate negative correlation with rainfall (r = -0.251 in Helsinki, r = -0.289 in Espoo). Rainfall exceeding 5.0 mm suppresses weekday ridership by 20%?35% relative to seasonal baselines.
? CRITICAL METHODOLOGICAL DISCLAIMER (Correlation != Causation): While warm, sunny days record high bike usage, Nordic summer temperature is strongly collinear with astronomical daylight (19 hours in June vs. 9 hours in October), university exam terms, and institutional summer shutdowns (July vacation). These empirical figures represent descriptive seasonal co-movements, not isolated causal elasticity.

6. DIURNAL PROFILES & THE COMMUTER EVENING RUSH SURGE | Operating-Day Normalized Intraday Regimes

Operating-Day Normalized Departure Dynamics (154 Weekdays vs. 60 Weekend Days per 214-day Season):
? Workday vs. Weekend Totals: Weekday volume exceeds weekend volume by +23.4% to +28.8% across both seasons. Helsinki: Weekday 10,708?10,985 trips/day vs. Weekend 8,530?8,679 trips/day; Espoo: Weekday 1,287?1,460 trips/day vs. Weekend 1,030?1,149 trips/day.
? Morphological Divergence: Weekdays follow a bimodal distribution with a sharp morning commute peak at 08:00 (~643?790 trips/hr in Helsinki) and an elongated evening commute plateau from 16:00 to 18:00 (~1,112?1,136 trips/hr). Weekends exhibit a smooth unimodal leisure curve (12:00?17:00).
? The Marked Evening Peak Surge (+70% to +105% Higher than Morning Peak):
- Helsinki: Morning (07:00?09:00) 1,208?1,287 trips/day vs. Evening (16:00?18:00) 2,154?2,194 trips/day (+70.5% in 2024; +78.4% in 2025).
- Espoo: Morning 141?150 trips/day vs. Evening 276?308 trips/day (+105.4% in 2024; +95.9% in 2025).
- Behavioral Driver: Morning trips are concentrated, direct home-to-work utilitarian journeys. Evening trips combine homeward commutes with dining, errands, waterfront leisure, and social trips, creating an expanded, dispersed travel peak.

Figure 5: Operating-Day Normalized Hourly Departure Dynamics: Weekday vs. Weekend
(Temporal Scope: 2024 & 2025 Seasons; 154 Weekdays & 60 Weekends per Season; Unit: Departures / Operating Day / Hour)

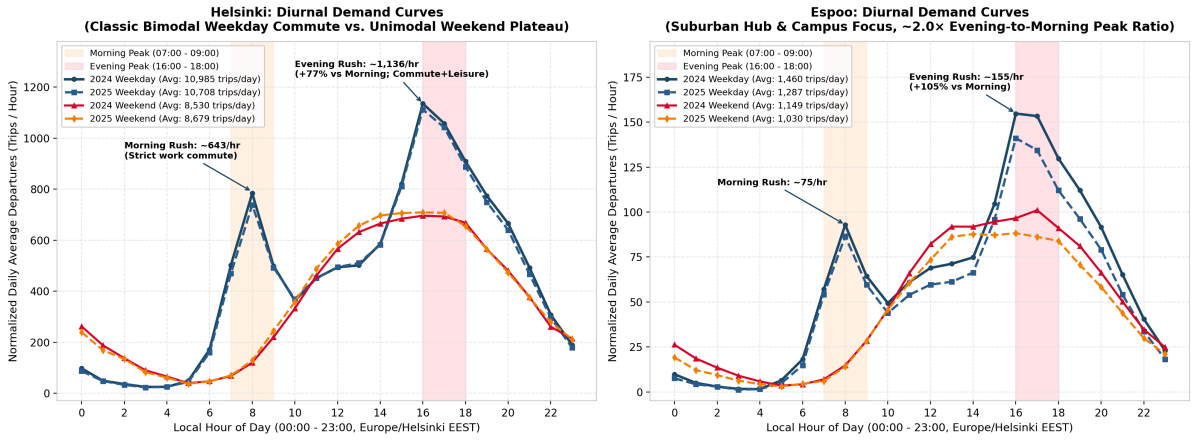


Figure 5: Operating-day normalized hourly departures (0-23h) across Helsinki (left) and Espoo (right). Contrasts bimodal weekday curves against unimodal weekend plateaus, highlighting the 70%-105% evening surge over morning rush.

7. CAUSAL WITHIN-SEASON MACHINE LEARNING FORMULATION | Solution B Operational Architecture

? Operational Decision Epoch: At the beginning of month m (May through October), fleet operators observe completed trip volume through month m-1. The task is to forecast completed monthly departures and returns for each station in month m to guide monthly rebalancing.
? Zero Lookahead Weather Guarantee: 2025 ground-truth weather was strictly excluded from predictive models. In subarctic Finland, long-range numerical weather predictions weeks in advance exhibit zero practical skill, and using actual historical weather would introduce fatal data leakage.
? Causal Feature Space (10 dimensions): Prior-month lag volume (lag_departures or lag_returns), station docking capacity, latitude, longitude, municipality indicator (is_helsinki), and month indicators (m_6 to m_10) to capture system-wide seasonal trajectory.

8. MULTI-MODEL BENCHMARK ON UNIFIED TEST COHORT | May·October 2025 (Identical N = 2,718 Station-Months)

Evaluation Dimension / Target Metric	1. Same-Month Baseline (2024)	2. Ridge Regression (?=10)	3. Decision Tree (depth=6)
Departures MAE (trips / station-month)	141.88 trips	226.01 trips	199.53 trips
Departures WAPE (%) - Volume Error	17.58%	28.00%	24.72%
Returns MAE (trips / station-month)	139.73 trips	226.52 trips	195.63 trips
Returns WAPE (%) - Volume Error	17.31%	28.06%	24.23%
Low-Demand Stations WAPE (Q1 <= 198 trips; N=681)	33.83% (MAE: 41.92)	124.07% (MAE: 153.74)	55.98% (MAE: 69.37)
Top-20 Net Deficit Overlap Range (Rank Order)	40% ? 75% (Avg: 60.8%)	60% ? 80% (Avg: 69.2%)	25% ? 40% (Avg: 31.7%)
Top-20 Net Surplus Overlap Range (Rank Order)	35% ? 75% (Avg: 53.3%)	60% ? 75% (Avg: 65.8%)	25% ? 35% (Avg: 30.8%)
Spearman Net Flow Rank Correlation (?) Range	0.483 ? 0.663 (Avg: 0.582)	0.455 ? 0.761 (Avg: 0.650)	0.123 ? 0.358 (Avg: 0.242)

Full-Season Baseline Reference Cohort (April?October 2025, N = 3,170 test station-months across all 7 months):
Departures MAE = 147.60 trips/month | WAPE = 19.23% | RMSE = 270.87 trips | Median Absolute Error = 73.0 trips | Pearson r = 0.959.
Returns MAE = 145.99 trips/month | WAPE = 19.02% | RMSE = 272.08 trips | Median Absolute Error = 71.0 trips | Pearson r = 0.959.

1. For Absolute Demand Volume: Historical Same-Month Baseline Achieves Lowest Error (17.58% vs. 28.00% Ridge vs. 24.72% Tree):
In Helsinki's subarctic climate, month-to-month transitions involve sharp seasonal step-changes (April thaw -> May warmup; August warmth -> September chill). Preceding-month lag models suffer from inherent transitional delay, whereas the prior year's same calendar month naturally aligns with solar irradiance, school term starts, and summer holiday schedules.
2. Severe Failure of Linear Models on Low-Demand Stations (Ridge WAPE = 124.07% on Q1 Docks):
On the quietest 25% of stations (Q1 <= 198 trips/month, N = 681), Ridge incurs a 124% WAPE error. Global month indicator coefficients apply rigid additive shifts across all stations, causing massive relative overprediction at peripheral docks. Baseline preserves localized scale (33.8% WAPE).
3. For Commuter Deficit Priority Screening: Ridge Regression Excels in Early/Mid-Season (Up to 80% Top-20 Overlap):
In May through August, Ridge achieves 60%?80% overlap on Top-20 deficit stations (Spearman ? up to 0.761). Its monotonic linear slope effectively magnifies heavy commuter sinks (e.g. Pasila). Planners should use Baseline for fleet sizing and Ridge for priority sink screening.

Figure 7: 2025 Out-of-Time Demand Forecasting Performance: Actual vs. Historical Same-Month Baseline
(Evaluated on Complete 2025 Season; Baseline Model: ŷ(s,m,2025) = y(s,m,2024); Unit: Trips / Station-Month)

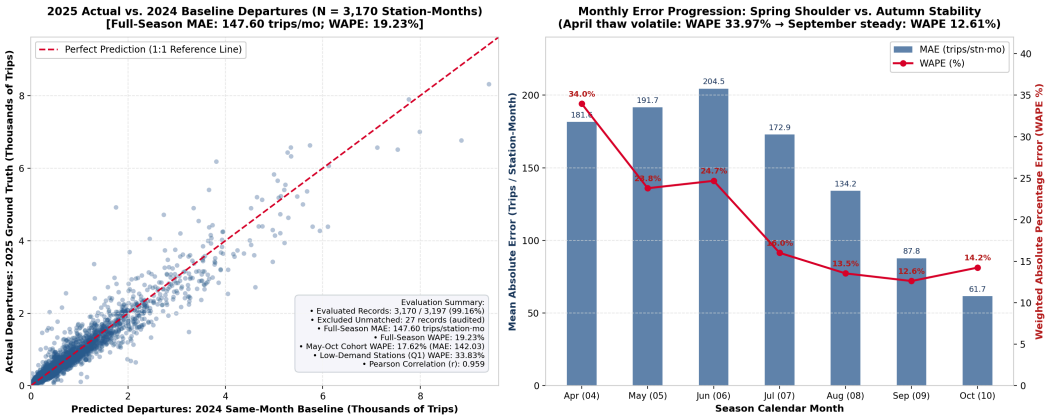


Figure 7: Out-of-time demand evaluation (2025 actual vs. same-month baseline). Left: Scatter against 1:1 diagonal (N=3,170, r=0.959). Right: Monthly MAE and WAPE progression showing spring transition variability and high autumn stability (September WAPE 12.61%).

9. SPATIAL POLARIZATION OF CUMULATIVE NET FLOW | Directional Imbalance Across 454 Mapped Stations

? Macro Net Flow Metric: net_flow = returns - departures calculated over all 14 operating months across 454 stations with coordinates.
? Systemic Deficit Sinks (Red, Bike Depletion Vulnerability): Concentrated around major rail interchanges and radial transit arterials where commuters arrive by train or bus and depart on city bikes, causing chronic docking shortages and bike depletion.
? Systemic Surplus Sources (Blue, Dock Congestion Vulnerability): Concentrated along scenic waterfronts, central commercial plazas, and dense residential docks where riders dock bikes upon arriving home or at leisure destinations, threatening return blocking.

10. STATION TRIAGE REBALANCING PRIORITY LISTS | Top 5 Chronic Deficit Sinks vs. Top 5 Chronic Surplus Sources

ID	Deficit Station (Depletion Sink)	City	Cap	Net Flow
113	Pasila railway station	Helsinki	40	-11,014 trips
120	Mäkelänkatu	Helsinki	28	-5,275 trips
203	Opastinsilta	Helsinki	40	-5,049 trips
043	Karhupuisto	Helsinki	24	-4,421 trips
162	Leppäsuonaukio	Helsinki	28	-3,750 trips

ID	Surplus Station (Dock Saturation)	City	Cap	Net Flow
011	Unioninkatu	Helsinki	22	+3,474 trips
012	Kanavaranta	Helsinki	34	+3,081 trips
125	Arielinkatu	Helsinki	16	+3,027 trips
404	Sompasaari	Helsinki	14	+2,900 trips
065	Hernesaarenranta	Helsinki	40	+2,572 trips

11. DIURNAL COMMUTER TIDAL REVERSAL | Peak Hour Asymmetry: Morning (07:00·09:00) vs. Evening (16:00·18:00)

Station Net Flow Reversal Across Standardized 2-Hour Weekday Peak Windows:
? Quadrant II: Residential Feeder Docks (Morning Deficit -> Evening Surplus):
Sompasaari (Station 404, Cap 14): In the morning rush, residents depart for work generating net flow of -7,973 trips (8,787 departures vs. 814 returns). In the evening rush, commuters return home, generating net flow of +2,808 trips. Total Diurnal Tidal Swing: 10,781 trips.
? Quadrant IV: Employment & Commercial Hub Docks (Morning Surplus -> Evening Deficit):
Salmisaarenranta (Station 059, Cap 28): In the morning rush, inbound workers dock bikes resulting in net flow of +7,343 trips (7,543 returns vs. 200 departures). In the evening rush, workers unlock bikes to leave offices, generating net flow of -5,244 trips. Total Diurnal Tidal Swing: 12,587 trips (#1 Tidal Hub).
Itämerentori (Station 030, Cap 40): Morning surplus +6,838 trips -> Evening deficit -5,022 trips (Total Diurnal Tidal Swing: 11,860 trips, #2 Tidal Hub).
? Quadrant III: Persistent Structural Sinks (All-Day Deficit):
Pasila railway station (Station 113, Cap 40): Remains in heavy deficit during both morning (-2,618) and evening (-2,607) peaks, acting as an all-day transit dispersal node.
? Pre-Staging Operational Hypothesis: Because departures surge sharply between 16:00 and 18:00, planners could explore an operational hypothesis of pre-staging bicycle redistribution prior to the afternoon peak (e.g., before 15:30); verifying this strategy requires real-time dock inventory data.

Figure 6: Morning Peak vs. Evening Peak Station Comparison: Revealing Diurnal Commuter Tidal Reversal
(Temporal Scope: 154 Weekdays Across 2024-2025; Morning: 07:00-09:00, Evening: 16:00-18:00; Unit: Net Flow Trips)

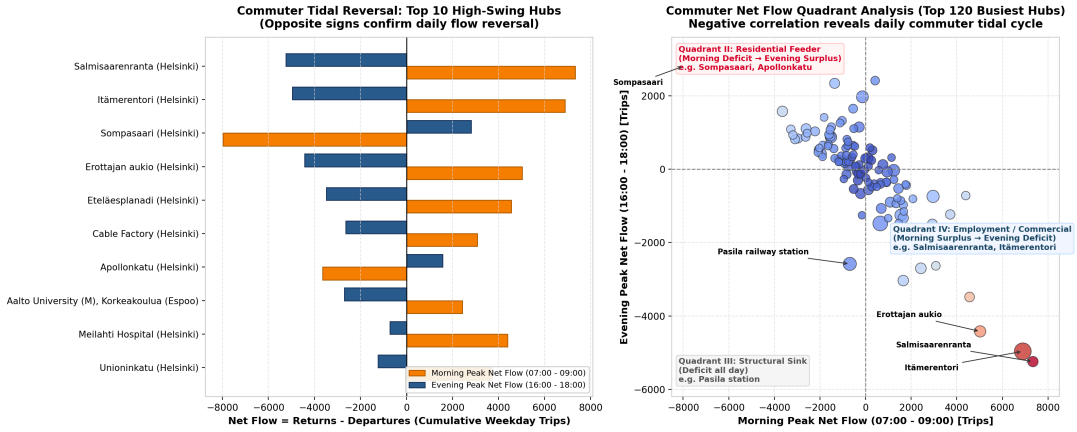


Figure 6: Station-level net flow during morning rush (07:00-09:00, x-axis) vs. evening rush (16:00-18:00, y-axis). Highlights diurnal tidal reversal: Residential feeders (Quadrant II), Commercial/employment hubs (Quadrant IV), and Structural sinks (Quadrant III).

12. REFLECTION: CANVAS PLAN VS. ACTUAL COMPLETED WORK | Methodological Evolution & Refinement

? Evolution of Weather Features (Decoupling Lookahead Bias): The Canvas Workshop 3 proposal initially contemplated using weather and local weather forecasts as direct regression features. However, during pipeline development, methodological auditing revealed that using ground-truth retrospective weather in an out-of-time forecasting benchmark creates fatal lookahead bias (future weather is unknown at decision time), while commercial numerical weather forecasts weeks/months in advance have zero skill in subarctic Finland. We correctly decoupled weather into retrospective descriptive EDA (identifying strong seasonal temperature correlation r ~ 0.74 alongside daylight/holiday confounders) and restricted predictive ML to causal lag-1 signals.

? Temporal Granularity & Operational Hierarchy: The Canvas proposed multi-scale regressions from hour to month. Modeling full-network hourly demands across 457 stations yields sparse, zero-inflated counts that obscure seasonal macro trends and require dock sensor telemetry. We structured a clean two-tier operational framework: (1) Monthly macro demand forecasting (Baseline vs. Ridge vs. Trees) for seasonal fleet sizing, and (2) Operating-day normalized diurnal curves (0-23h) and peak-hour net flow reversal for day-to-day dispatch planning.

? Metric & Task Differentiation: The Canvas envisioned general allocation; our empirical evaluation proved that volume sizing (MAE/WAPE) and rebalancing triage (Spearman rank correlation) require different tools: Historical Baseline for sizing vs. Ridge for early-season deficit screening.

13. METHODOLOGICAL CONSTRAINTS & PHYSICAL REALITY BOUNDARIES | Analytical Disclaimers

1. Historical Weather != Weather Forecasts:
FMI weather records represent retrospective ground-truth observations, strictly excluded from predictive models to eliminate data leakage.
2. Descriptive Correlation != Causation:
Daily trip correlations (r ~ +0.74 with temp) reflect seasonal co-movement. Astronomical daylight (19h June vs. 9h Oct), school calendars, and summer holidays are heavily collinear with temperature.
3. Forecast Error Unit (MAE) != Physical Bicycles:
MAE is measured in trips per station-month (e.g. 147.60 trips/mo). It does not represent physical bicycle shortages; one bicycle serves multiple trips daily.
4. Monthly Net Flow != Real-Time Van Dispatch Workload:
Cumulative monthly net_flow = returns - departures reflects 30-day passenger drift, not immediate truck dispatch needs. Diurnal tidal reversals partially offset daytime imbalances; truck dispatch requires 15-minute dock occupancy sensor telemetry.
5. Single Out-of-Time Test Cycle (2025 Season):
Evaluating on 2025 provides an exploratory out-of-time backtest, not multi-year statistical significance.
6. Partial Q4 Operating Window:
The HSL season closes October 31; Q4 contains only October (1 operating month).

14. DATA PRIVACY, ETHICAL GOVERNANCE & LICENSING | Compliance Framework

? Data Privacy Protection: In strict compliance with European GDPR and ethical research standards, all analytical tables and published web deliverables aggregate trips at station-month or hourly municipal scales. Zero individual rider identifiers, trajectory coordinates, or personal smart-card tokens exist in raw or derived data.
? Attribution & Open Data Licensing: Ingested journey records and ArcGIS station metadata are published by City Bike Finland and HSL under Creative Commons Attribution 4.0 International (CC BY 4.0). Weather observations are provided by the Finnish Meteorological Institute (FMI) under CC BY 4.0.

15. KEY LEARNING TAKEAWAYS & FUTURE IMPROVEMENTS | Engineering Synthesis

Key Project Learning Takeaways:
1. Domain Physics Beat Algorithmic Complexity: In subarctic shared mobility, seasonal calendar alignment (same-month baseline) outperforms complex feature regressions that suffer from transitional lag across Nordic seasons.
2. Strict Leakage Auditing is Paramount: Decoupling weather lookahead prevented deceptive model performance and yielded realistic operational benchmarks.
3. Multi-Metric Evaluation: Single-metric optimization (e.g. global MAE) hides severe localized failure modes, such as Ridge's 124% WAPE on quiet stations.

Future Operational Improvements:
1. 15-Minute Sensor Telemetry: Ingest real-time dock occupancy GBFS feeds to transition from macro seasonal triage to dynamic rebalancing truck routing.
2. Same-Day Weather Adjustment Heuristics: Develop day-of-forecast multiplier adjustments based on short-range 24h FMI precipitation forecasts.
3. User Crowdsourced Rebalancing Incentives: Model dynamic reverse-pricing or bonus credits for users riding from surplus docks to deficit sinks.

Interactive Planning Web Application (GitHub Pages Platform):
Delivered as a zero-dependency static web platform hosted under docs/ (published to GitHub Pages at https://luosuo890.github.io/HSL_City_Bike_Project/). Features 4 interactive modules: (1) Leaflet Station Map with net flow & capacity layers; (2) Intraday Commute & Tidal Peaks; (3) FMI Daily Weather Correlation Explorer; (4) Model Comparison & 2025 Test Predictions Scatter. 22 automated test suites passing.